Aidan Altorelli

Allocation LLM Schema

What the agent will do:

1. Receive client profile from the Profile Agent and a market view/asset view from the Research Agent
2. Produce a target portfolio including weights across asset classes, factor tilts, and specific instruments.

Agent Design:

Inputs:

- Structured Client profile from Profile Agent (age, income, savings, illiquid and liquid assets etc.) (Assumed inputs Agent dependent)
- Market View from Research Agent (expected returns by asset class, covariance matrix, regime indicators) (Assumed inputs Agent dependent)
- Universe of allowable instruments in the portfolio

Outputs:

- Target weights → List of every instrument the client should hold and how much investable capital goes into each. Should sum to 100%. Only pre-approved assets from the allowable instrument input.
- Portfolio Split → the share of the portfolio in risky assets and safe assets
- Explanation → short explanation of the position and why it is held at that weight

- Decomposition of each weight →. A breakdown showing how much of each position comes from (a) the market-equilibrium baseline, (b) tilts driven by the Research agent's views, and (c) adjustments made to offset human capital exposure.
- Expected portfolio statistics → The agent's estimate of the portfolio's expected return, volatility, and key factor exposures

Expected Failures:

- Hallucinated Tickers or non-existent funds
- Concentration in single names the LLM "likes", enforce position limits programmatically, not via prompt.
- Inconsistent reasoning vs. weights (rationale says "underweight tech" but weight is 25%), add a self-consistency check.
- Over-reliance on trailing returns mentioned in the prompt, prompt the model with forward-looking inputs only.
- LLM picking exotic strategies to seem sophisticated (the AUM-complexity problem replicated by an AI), constrain the universe.

Validation Plan:

- Backtest the three personas over rolling windows; compare Sharpe and drawdown vs. naive 60/40 and target-date funds.
- Unit test: for the tech executive persona with concentrated RSUs, verify the financial portfolio's tech weight is materially lower than the biology professor's.
- Stability test: run the same persona 20 times, measure dispersion in weights. If allocations swing wildly across runs, the agent isn't deterministic enough.

Risk LLM Schema

What the agent will do:

1. Receive the proposed portfolio from the Allocation Agent and the client profile from the Profile Agent (including human capital characteristics).
2. Produce a risk assessment of the portfolio standalone, on a total-balance-sheet basis (portfolio + human capital), under stress scenarios, and against fiduciary standards. Output a decision (approve, flag, reject) with reasoning.

Agent Design:

Inputs:

- Proposed portfolio from Allocation Agent (target weights, risky/safe split, expected statistics) *(Assumed inputs Agent dependent)*
- Client profile from Profile Agent including human capital model: PV of future earnings, factor exposures, employer concentration *(Assumed inputs Agent dependent)*
- Historical factor returns and pre-defined stress scenarios (2008 GFC, 2020 COVID, 1970s inflation, employer shock)
- Fiduciary and suitability constraints (objectives, liquidity, time horizon)

Outputs:

- Decision → APPROVE, FLAG, or REJECT.
- Portfolio-level metrics → Standalone statistics on the financial portfolio: volatility, VaR/CVaR at 95% and 99%, max drawdown, factor exposures, liquidity score.

- Human-capital-adjusted metrics → Risk statistics on the total balance sheet (portfolio + human capital): total equity beta, sector concentration, share of total wealth at risk. Core contribution of the project.
- Scenario stress results → Impact on portfolio and total wealth under each scenario. The employer shock (RSUs to zero + job loss) is critical for the tech executive persona.
- Flags → Specific issues with severity and suggested fix.
- Fiduciary notes → Whether the portfolio meets objectives, fees, liquidity, and suitability.
- Reasoning trace → Structured explanation citing specific metrics for auditability.

Expected Failures:

- Rubber-stamping Allocation due to cross-agent sycophancy, mitigate with adversarial prompt framing.
- Hallucinated quantitative outputs, compute all numerical metrics deterministically in code; the LLM interprets but does not generate numbers.
- Over-confidence in human capital estimates, report across a range of assumptions rather than a point estimate.
- Missing tail risks not in training data, anchor stress tests to historical factor returns.
- Inconsistent flagging across runs, use temperature 0 and self-consistency checks.

Validation Plan:

- Adversarial test set: construct clearly bad portfolios (100% employer stock, leveraged ETFs for a retiree) and verify near-100% rejection.
- Human capital sensitivity: verify the tech executive's HC-adjusted total equity beta is materially higher than standalone; opposite for the biology professor with a pension.
- Cross-agent disagreement metric: measure how often Risk flags or rejects Allocation proposals. Near zero indicates rubber-stamping; a healthy middle range demonstrates independence.
- Stability test: run the same portfolio 20 times. Decision and flags should be consistent across runs.